Wadi Boughanmi

Geomatics Engineer | Full-Stack WebGIS Developer

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PROFESSIONAL PROFILE

Geomatics engineer specializing in geographic information systems and databases. Excellent knowledge of all stages of cartographic production. Solid experience in the development of 2D and 3D mapping web solutions, automation of GIS workflows with ArcPy, and geospatial AI.

SKILLS

Geomatics software	ArcGIS, ArcGIS Pro, ArcGIS Enterprise, QGIS, Global Mapper, Agisoft Metashape
GIS and automation	ArcPy (Esri), PostgreSQL, PostGIS, geodatabase design (Enterprise Architect), OGC services (WMS, WFS), spatial analysis
Remote sensing	Satellite image analysis, Sentinel-2, Google Earth Engine, change detection
Cartography	Topographical and thematic maps, cartographic generalisation
Web development	Django, GeoDjango, Django REST Framework, Leaflet, OpenLayers, Cesium, Three.js, GeoServer, Turf.js, WAMP, NGINX
Programming	Python, JavaScript, HTML, CSS, Flutter
AI / deep learning	TensorFlow, Keras, PyTorch, YOLO, DeepSort
Personal skills	Project management, teamwork, communication

PROFESSIONAL EXPERIENCE

Geomatics Engineer | Full-Stack WebGIS Developer
National Center for Cartography and Remote Sensing (CNCT) | August 2022 – Present

- Architect and build enterprise WebGIS platforms end to end, from PostGIS data models and Django REST APIs to interactive Leaflet, OpenLayers and Cesium front ends.
- Publish and secure OGC services (WMS, WFS) with GeoServer to feed 2D and 3D web maps with live spatial data.
- Ship and maintain production geospatial applications for external stakeholders under strict confidentiality requirements.
- Automate recurring GIS workflows with ArcPy and Python, replacing manual map production and data processing steps.
- Lead the GIS department: plan projects, set priorities and guide the team through delivery of geospatial software.

INTERNSHIPS

Application Internship | CNCT (January 2022 - June 2022)
- Development of a web and Android GIS for thematic mapping.

PFE internship | CRM (February 2021 - June 2021)
- Preparation of a multi-class dataset for the detection of small objects using a Deep Learning approach.

PFA internship | EABA (February 2020 - June 2020)
- Display radar data on a digital map in real time.

Summer internship | CNCT (July 2019 - August 2019)
- Monitoring of the cartographic production chain.

EDUCATION

Degree in Engineering (2021)
Engineering cycle in Geomatics | Borj El Amri Aviation School (EABA)

Preparatory cycle diploma (2018)
Preparatory engineering cycle | EABA

Baccalaureate, Highest Honors (2016)

PROJECTS

Urban Awareness (Synoptica)
Interactive Tunisia dashboard that scores access to health, security and transport services, adjusts the score for population pressure, simulates new facilities, maps service gaps and routes to the nearest service. Shows live air quality and weather.
Stack: Leaflet, Turf.js, spatial analysis, MarkerCluster, Routing Machine, heatmaps

Urban Analysis GIS: shortest-path engine
Calculates shortest paths in the browser with Dijkstra's algorithm on a graph pre-computed in PostGIS, limited to a polygon zone and run with Web Workers. Includes nearest and nearby analysis of points of interest across Tunisia.
Stack: PostGIS, Dijkstra, Web Workers, Leaflet

GeoTime Machine
Interactive time-series tool (2018-2025) to monitor environmental and urban evolution: compares historical satellite imagery, tracks vegetation health (NDVI) and built-up area (NDBI) with annual trends, and quantifies vegetation gain/loss and urban expansion inside user-defined areas of interest.
Stack: Leaflet, Google Earth Engine

3D Terrain Intelligence
3D application built with CesiumJS to explore and analyse Tunisia's terrain: elevation inspection, slope and aspect analysis, elevation profiles, hillshade, line-of-sight analysis and water-level simulation.
Stack: Cesium, AWS Terrain Tiles

MyCarCare: car maintenance and roadside assistance
Platform for car owners in Tunisia: tracks maintenance and paperwork deadlines, sends emergency requests to nearby service providers, shows the provider's live position on a map, with bookings, ratings and an AI chatbot.
Stack: Django REST Framework, Channels, SSE, Leaflet, Gemini

Automatic cartographic generalisation
Automates map generalisation in ArcGIS with ArcPy: simplifying and aggregating features to produce clean maps at smaller scales.
Stack: ArcPy, ArcGIS, Python

Change detection
Detects land-cover change from Sentinel-2 imagery processed in Google Earth Engine, with results shown on a Leaflet map served by Django.
Stack: Google Earth Engine, Sentinel-2, Django, Leaflet

3D mapping app
Web platform to search, query and analyse spatial data in 3D.
Stack: Cesium, PostgreSQL, PostGIS, GeoServer

2D mapping app
Web map to query spatial data and read clear statistics from it.
Stack: GeoDjango, Leaflet, GeoServer

Object detection and tracking
Detects and tracks objects in drone footage using YOLO with DeepSort multi-object tracking.
Stack: Python, YOLO, DeepSort

Task management system
Fully functional task management system.
Stack: Django, Chart.js

TRAINING COURSES

- Scrum Fundamentals for Scrum Master and Agile Projects - Udemy
- Fundamentals of Network Security - Udemy
- Computer Vision with Embedded Machine Learning - Edge Impulse

ARTICLE

An Upgraded-YOLO with Object Augmentation: Mini-UAV Detection Under Low-Visibility Conditions by Improving Deep Neural Networks (30/09/2022).

LANGUAGES

Arabic (native) | French (C1) | English (C1)